

PALLADIUM BANK — RETAIL ANALYTICS STAR SCHEMA DIAGRAM

PALLADIUM BANK — RETAIL ANALYTICS STAR SCHEMA

Dimensional Data Model | PostgreSQL | Grain: Per Transaction

dim_date [DIMENSION]	
date_key [PK]	INT
-- YYYYMMDD surrogate	
full_date	DATE
day_of_week	TINYINT
month_number	TINYINT
month_name	VARCHAR
quarter_number	TINYINT
quarter_label	CHAR(2)
-- Q1_Q4	
year_number	SMALLINT
year_month	VARCHAR
-- 2024-01	
year_quarter	VARCHAR
-- 2024-Q1	
is_weekend	BOOLEAN
fiscal_year	SMALLINT

dim_customer [SCD TYPE 2]	
customer_sk [PK]	BIGINT
-- Surrogate key	
customer_id [NK]	VARCHAR
-- Business key	
customer_name	VARCHAR
tier	VARCHAR
-- Platinum/Gold/Silver/Standard	
tier_start_date	DATE
-- SCD2 start	
tier_end_date	DATE
-- NULL = current	
is_current	BOOLEAN
-- Latest version flag	
home_branch_id	VARCHAR
home_state	VARCHAR
account_open_date	DATE
customer_tenure_months	INT

dim_channel [DIMENSION]	
channel_sk [PK]	INT
-- Surrogate key	
channel_name	VARCHAR
-- POS/ATM/Branch/USSD/...	
channel_type	VARCHAR
-- Physical/Digital/Self-Svc	
is_digital	BOOLEAN
is_self_service	BOOLEAN

fact_transactions [FACT Grain: per transaction]	
txn_fact_sk [PK]	BIGINT
-- Surrogate key	
txn_id [DegDim]	VARCHAR
-- Degenerate dimension	
date_key [FK]	INT
--> dim_date	
customer_sk [FK]	BIGINT
--> dim_customer (SCD2)	
product_sk [FK]	INT
--> dim_product (SCD1)	
branch_sk [FK]	INT
--> dim_branch (SCD1)	
channel_sk [FK]	INT
--> dim_channel	
txn_type_sk [FK]	INT
--> dim_txn_type	
txn_amount	NUMERIC
-- Transaction amount (NGN)	
balance_after	NUMERIC
-- Balance after txn	
fee_amount	NUMERIC
-- Fee income	
txn_hour	TINYINT
-- Hour 0-23	
txn_timestamp	TIMESTAMP
-- Full datetime	

dim_product [SCD TYPE 1]	
product_sk [PK]	INT
-- Surrogate key	
product_id [NK]	VARCHAR
-- Business key	
product_name	VARCHAR
product_type	VARCHAR
-- Account/Card/Loan/Digital	
is_deposit_product	BOOLEAN
is_withdrawal_product	BOOLEAN
is_digital_product	BOOLEAN
is_active	BOOLEAN

dim_branch [SCD TYPE 1]	
branch_sk [PK]	INT
-- Surrogate key	
branch_id [NK]	VARCHAR
-- Business key	
branch_name	VARCHAR
state	VARCHAR
-- Lagos/Abuja/Kano/Rivers	
city	VARCHAR
region	VARCHAR
is_active	BOOLEAN
opened_date	DATE

agg_monthly_branch_summary [AGGREGATE TABLE]	
summary_sk [PK]	BIGINT
-- Surrogate key	
year_month	CHAR(7)
-- 2024-01	
branch_sk [FK]	INT
--> dim_branch	
branch_name	VARCHAR
state	VARCHAR
total_transactions	INT
unique_customers	INT
total_txn_amount	NUMERIC
total_fee_income	NUMERIC
total_deposits	NUMERIC
total_withdrawals	NUMERIC

dim_transaction_type [DIMENSION]	
txn_type_sk [PK]	INT
-- Surrogate key	
txn_type_name	VARCHAR
-- Deposit/Transfer/POS Purchase	
txn_direction	VARCHAR
-- Inflow/Outflow/Neutral	
generates_fee	BOOLEAN
fee_category	VARCHAR

Fact Table SCD Type 2 Dimension SCD Type 1 Dimension Static Dimension Aggregate Table